

Replication Project

Applied Stats II



31st March 2026

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Research Question & Theory

Replication and Extension of Jealousy Responses to Rival Characteristics

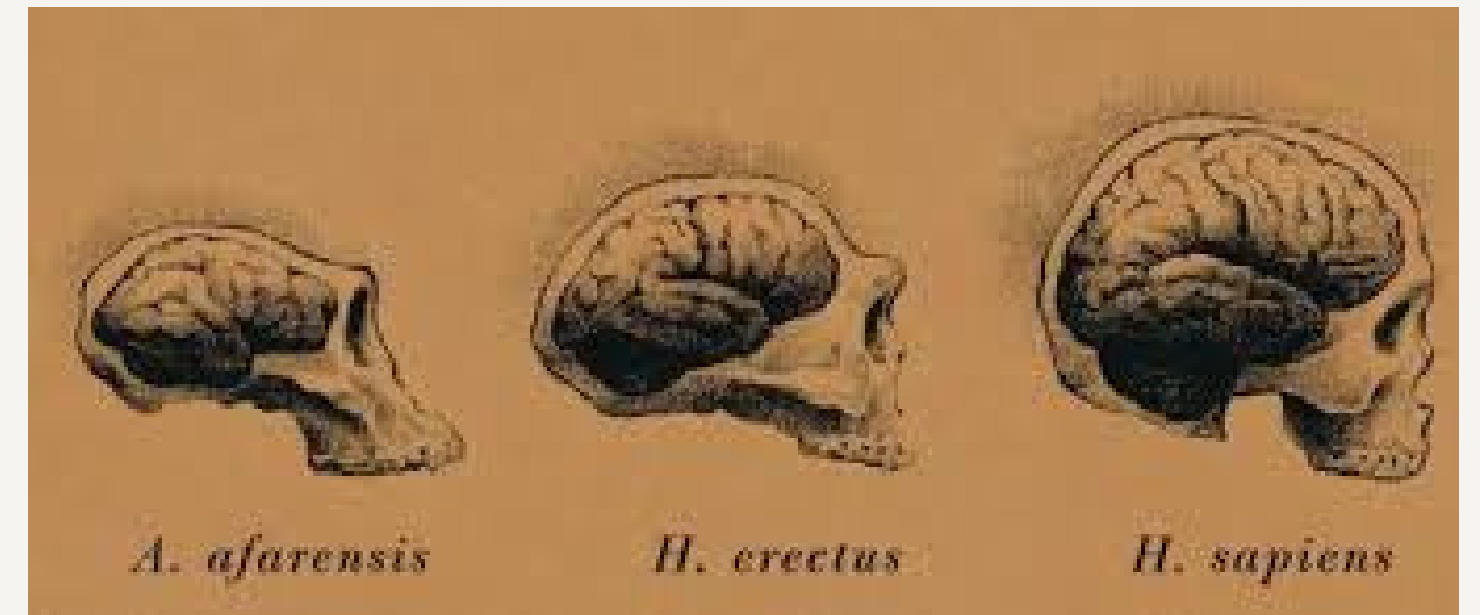
- Replicating a 2020 evolutionary psychology study
- Focus on jealousy responses to rival attractiveness
- Two extensions: robustness + model improvement

Research question:

- Do men and women respond differently to rival attractiveness?

Theoretical expectation:

- Women → more sensitive to attractiveness
- Men → more sensitive to dominance



Data & Empirical Setup



- 01
 - Two studies:
 - Study 1: 339 observations
 - Study 2: 456 observations
 - Sample: participants in relationships
- 02
 - Dependent variable: jealousy (ordinal scale)
 - Key variables:
 - Attractive condition
 - Gender
 - Dominance (control)
- 03

I replicate a study on jealousy and extend it by testing robustness across samples and using a more appropriate statistical model.
- 04

The key idea is that men and women may respond differently to different types of threats, and here we focus on attractiveness.
- 05

The dataset combines two samples and uses experimental conditions to isolate the effect of attractiveness.

Replication Strategy & Model

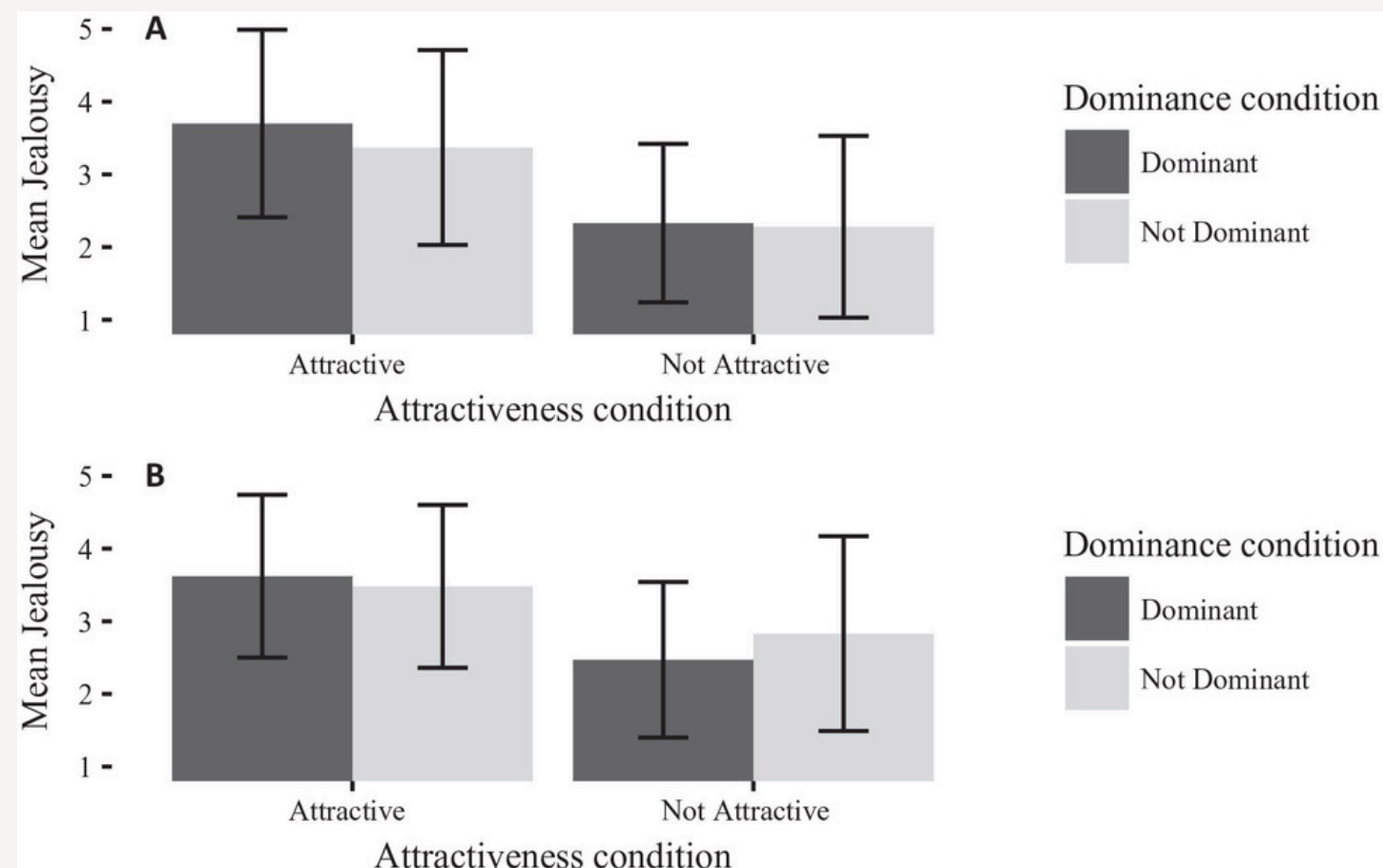
- Original method: factorial ANOVA

Bullet points:

- Model includes:
 - Attractiveness
 - Gender
 - Dominance
 - All interactions
- Focus: Attractive × Gender interaction

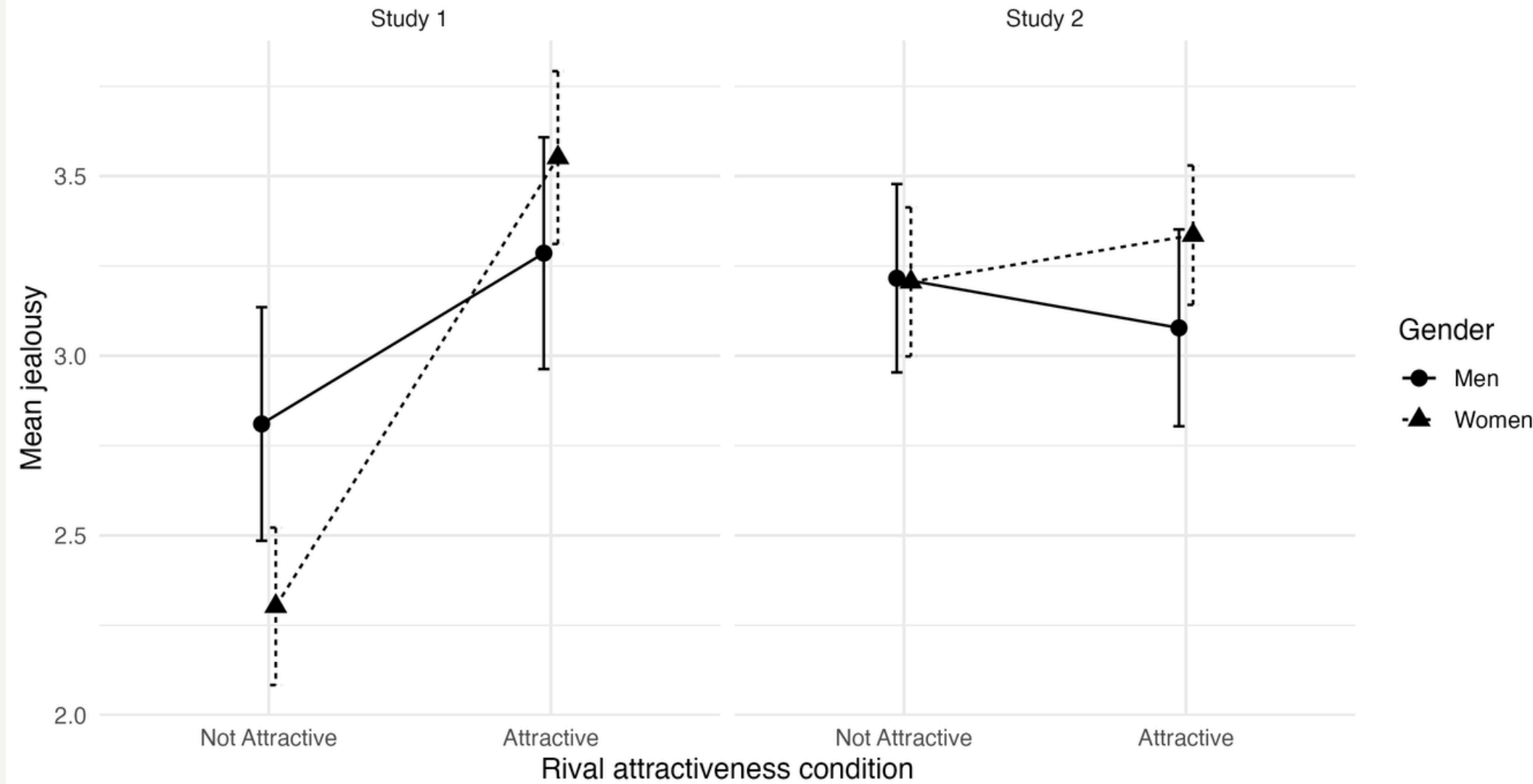
I replicate the exact model from the original study and focus on whether attractiveness affects men and women differently.

The figures here are the results of the original study.



For the scope of this project, the results for male jealousy and dominance has been omitted as there was no conclusive evidence for the dominance based evolutionary theory in this study.

Jealousy by rival attractiveness, gender, and study



Main Replication Results

- Study 1:
 - Strong increase in jealousy for women
 - Smaller increase for men
- Study 2:
 - Weak or flat pattern for both

In Study 1, women's jealousy increases sharply when the rival is attractive, much more than men's.

In Study 2, that pattern disappears — both genders show very little change.

This already suggests the effect is not consistent.

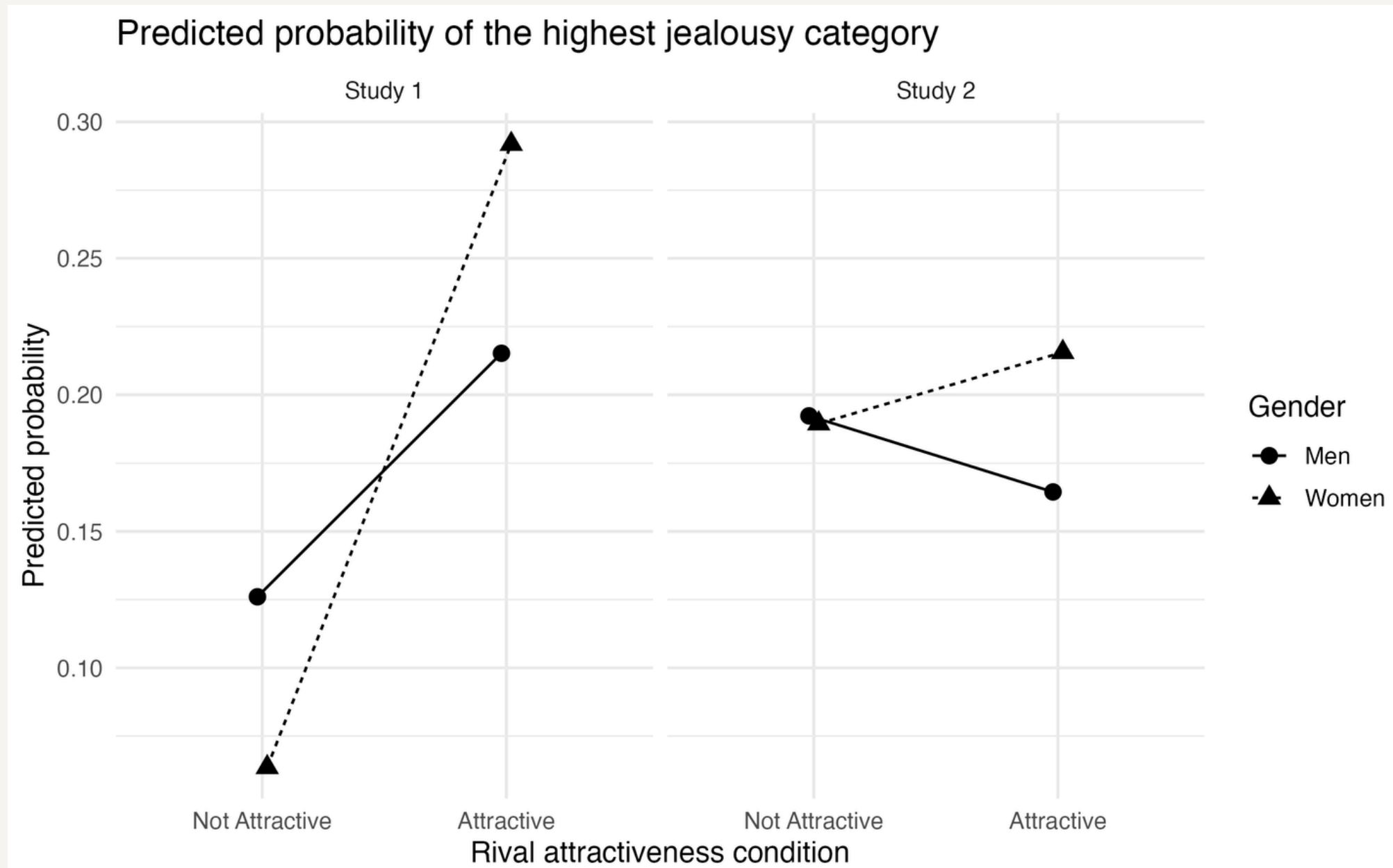
Cross-Study Robustness

Compare same interaction across both studies

- Strong effect in Study 1
- Weak or absent in Study 2
- Indicates lack of robustness

This extension shows the main result depends heavily on the sample — it is not stable across studies.

Extension 2: Ordered Logit



The key reason for using an ordered logit model is that jealousy is not truly continuous — it is measured on an ordinal scale, like a Likert scale.

In a linear model, we treat the difference between values like 1 and 2 the same as between 4 and 5, but that may not reflect how people actually respond.

The ordered logit model instead respects the ranking of the categories and models the probability of being in higher levels of jealousy.

This is particularly useful because it captures extreme responses, which are often more important in behavioural outcomes like jealousy.

So overall, this model improves both the statistical validity and the interpretation of the results.

- Ordered logit model (better for ordinal data)
- Study 1:
 - Large increase in high jealousy probability
- Study 2:
 - Small or inconsistent change

Using a better model, we see the same pattern: a strong effect in Study 1, but not in Study 2.

So the conclusion does not depend on the model — but the inconsistency remains.

01

Main findings

- Evidence for Attractiveness × Gender effect in Study 1
- Effect is weak or absent in Study 2
- Partial replication

02

What we learn

- Effect is not robust across samples
- Likely context- or sample-dependent
- Not a universal gender difference

03

Methodological contribution

- Cross-study comparison → tests robustness
- Ordered logit → appropriate for ordinal outcomes
- Results consistent across model

04

So the main takeaway is that while the theoretical prediction has some support, it is not consistently observed across studies.

05

This project shows that empirical results should be interpreted as conditional and context-dependent, rather than universally true.

Conclusion & Takeaways



Recent Developments in this area:

Article | [Open access](#) | Published: 17 December 2025

Facial femininity of potential rivals predicts jealousy in both heterosexual and lesbian women

[Junzhi Dong](#), [Benedict C. Jones](#) , [Esperanza Miyake](#) & [Victor K. M. Shiramizu](#)

[Scientific Reports](#) **15**, Article number: 43997 (2025) | [Cite this article](#)

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